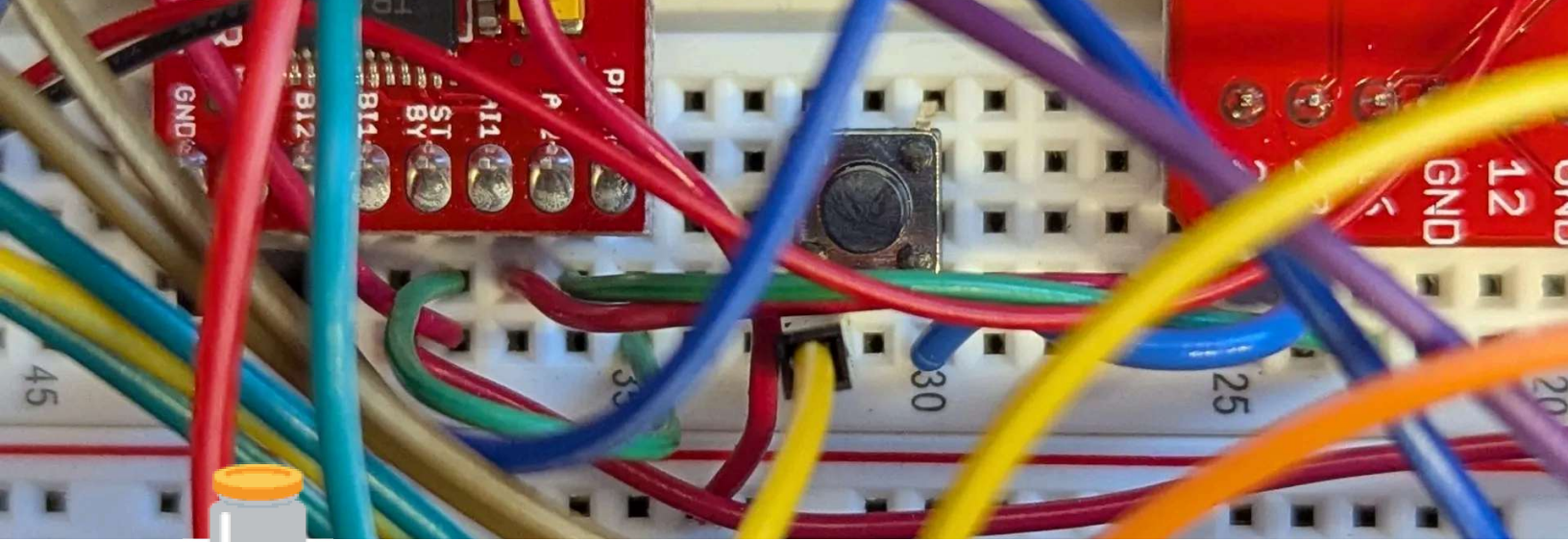




Design Project 3: Pill Bottle Opener

Overview



Skills Applied/Learned

✓ Improved Python skills

✓ GPIOZero Library + Raspberry Pi

✓ Wiring and soldering

✓ Breadboard

✓ Interpersonal skills

✓ Communication: Oral and Written

✓ Collaboration: Working in sub teams

✓ Time Management



Background Info:

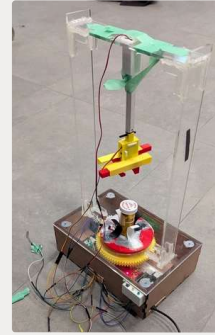
- **The child proof cap dilemma:** Must prevent children from opening bottles, while somehow allowing seniors with limited hand dexterity to as well.
 - this is often an impossible task because senior hand dexterity can be **similar** to that of children



Our idea for a solution

- Create a machine that can **automatically** open bottles with very **limited input**
 - will **push down, twist** and **lift** the lid off a pill bottle
 - pills can then be taken out **without struggle** with lid

Our final prototype



▼ Design Process



Insights + Research

- Age related conditions like arthritis are associated with **reduced grip strength** and **impaired fine motor control**, making repetitive hand tasks more difficult
- **14%** of **over 81 year old's** were unable to open a screw cap bottle
- Child resistant bottles are used to prevent children from **accidental** exposure or ingestion of medication
- Anything that is harmful when ingested is **required** to have a **child-resistant** cap by **law**



Identifying Goals: the problem to solve



▼ Objectives

- Should be **conveniently fast**
- Should be **easy** to use **without much dexterity** required
- Should require **minimal** manual effort from consumer



▼ Constraints

- Must **secure** medicine bottle while in use
- Must **open** bottle
- Must conform to **health and safety standards**
- Must **close** bottle



My Initial Ideas

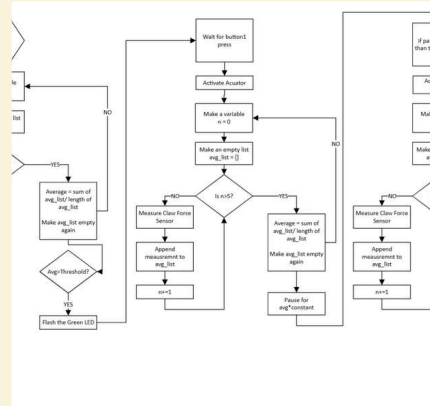
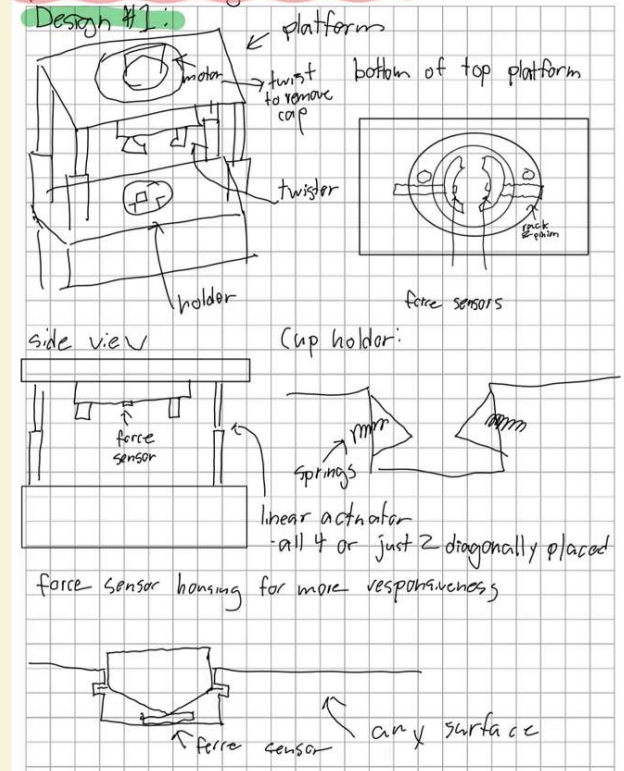
- **Device:**

- an box style case
- 4 **linear actuators** in the poles to raise and lower the platform
- A rotating base on the top platform that can rotate with a **gear** mechanism
- On the base will be a cylinder containing a **"gripper"** mechanism to grasp a lid, this would be a **rack and pinion** mechanism
- The rotating base would **twist** the lid after the platform lower **squeezes** it down
- A "cup holder" to **grip** and fix the bottle in place
- LEDs to signal status of the device
 - red, yellow and green
- Force sensitive resistors (FSRs) to detect the **presence** of the cup, that the lid is sufficiently **pressed** and the lid is **untwisted**

My rough

sketches/flowchart

David Wang wangd202 400620358





Brainstorm and discussion with group

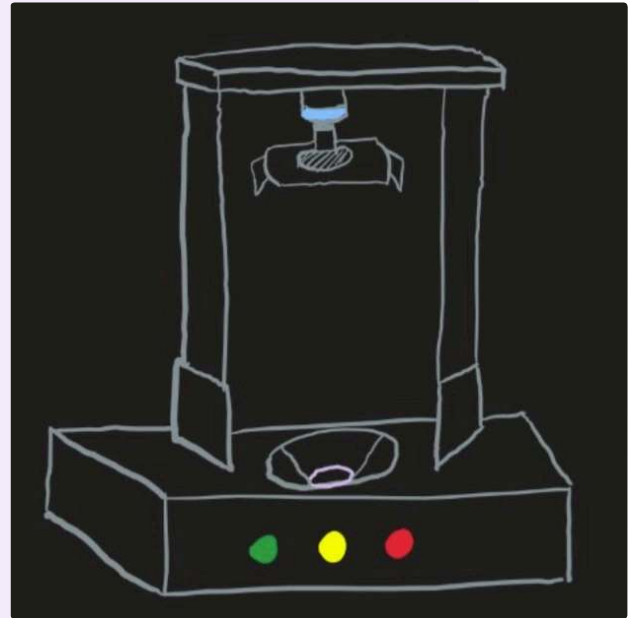
Device:

- Due to **limitations** with the raspberry pi, 4 linear actuators was **too many** However we could still do it with **just one**.
- It would be easier to **only** move the gripper rather than the whole platform, we switch to a design more like a press.
- The rack and pinion mechanism for gripping was decided to be **good**.
- agreed on the cup holder for the bottom base.

Code & wiring:

- We agreed to **proceed** with the plan we had, and to decide what to change **after testing**.

Our whiteboard notes/Sketches from discussion



some **edge cases** yet to be discovered.



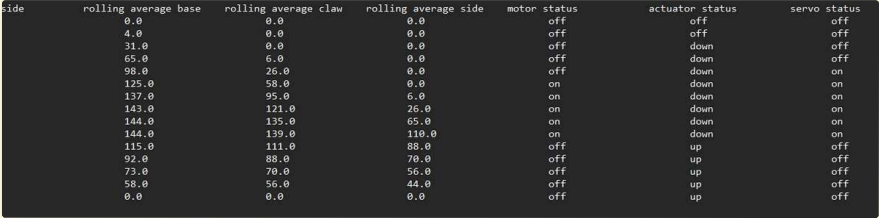
▼ Prototyping

In the Computing sub-team we began to **write a python program** to control the electronic components of the device

Wiring the breadboard and components was also done, some soldering was required. Later we organized the wires using electrical tape.

1 First Major Program

- We established the **basic** functionality. A foundational while loop constantly **reads** the force sensors and **triggers** output motors, actuators and LEDs. The inputs are run through functions that calculates initial and subsequent **rolling averages**.



side	rolling average base	rolling average claw	rolling average side	motor status	actuator status	servo status
0.0	0.0	0.0	0.0	off	off	off
4.0	0.0	0.0	0.0	off	off	off
31.0	0.0	0.0	0.0	off	down	off
65.0	6.0	0.0	0.0	off	down	off
98.0	26.0	0.0	0.0	off	down	on
125.0	58.0	0.0	0.0	on	down	on
137.0	95.0	6.0	0.0	on	down	on
143.0	121.0	26.0	0.0	on	down	on
144.0	135.0	65.0	0.0	on	down	on
144.0	139.0	110.0	0.0	on	down	on
115.0	111.0	88.0	0.0	off	up	off
92.0	88.0	70.0	0.0	off	up	off
73.0	70.0	56.0	0.0	off	up	off
58.0	56.0	44.0	0.0	off	up	off
0.0	0.0	0.0	0.0	off	up	off

image of text file output showing state of parts

➡ Changes to make

- Thresholds for the force sensors are **arbitrary** and placeholders. Output text is outputted to a **text file**. It is hard to read, also inefficient as we have it opening and closing with each loop. There are also **formatting** errors in the text.
- There was an error where the **wrong** outputs were tied to the wrong force sensors.
- The LED outputs are very simple, not very descriptive.



▼ Features

Demo Video

▼ Conclusion + Reflection

Place the bottle in



Team20 DP3 Fun



My Overall Contributions

- Brainstorming, ideation, design
- **Wired** much of the breadboard and components
- Contributed half of the **Python code**
- Coordinated **3D printing** of the parts
- Help with assembly of the device

close around the lid



What Went Well (Expand for details)

▼ Scope

I'm glad that we have gone for a **more complex** this time around. This project really **challenged** all of us to learn **new** things to accomplish what we eventually built.

▼ Communication and Dynamic

I got along very well this team. It was an **enjoyable** to work with a team I was **excited**



Challenges and Lessons

▼ Time and Pacing

We were much better at consistent work on the project and managing time well, however, we **underestimated** how many challenges we would run into and the time it took to fix. It became a slight crunch in the end. In the future, a **sizable** amount of challenges propping up in the design process will be expected,